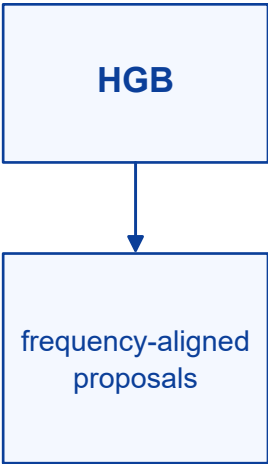


Dual-mode provenance triage for exploratory replacement proposals

A. Dual-mode proposal routes

Conservative Mode

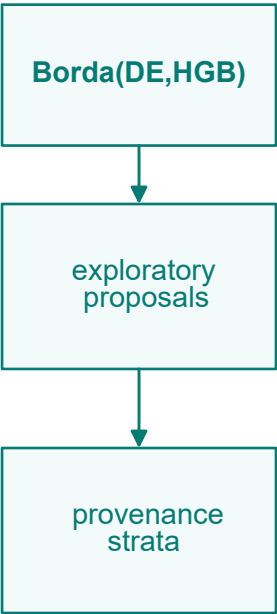
HGB route



Driven by HGB

Exploration Mode

Borda(DE,HGB) route



Driven by Borda(DE,HGB)

B. Provenance strata

G4 shared

$$K_{\text{HGB}} \cap K_{\text{Borda}}$$

consensus

G3 DE-elevated

$$K_{\text{Borda}} \setminus K_{\text{HGB}}$$

similarity-supported expansion

G2 Borda-emergent

$$K_{\text{Borda}} \setminus (K_{\text{HGB}} \cup K_{\text{DE}})$$

highest novelty

C. Coverage-limited A4C triage

| Group | Coverage                                              | Alert signal                                  |
|-------|-------------------------------------------------------|-----------------------------------------------|
| G2    | 100% <div><div></div></div>                           | 46.85%                                        |
| G3    | 100% <div><div></div></div>                           | 9.67%                                         |
| G4    | 5.63% <div><div></div><div>coveredunknown</div></div> | 17.60%<br>among covered;<br>94.37%<br>unknown |

A4C = computational triage only;  
not experimental validation.